



AB162x Series EVK Layout Guide

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Document Revision History

Revision	Date	Description
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1 Introduction

The AB1625 / AB1623 is an advanced single-chip solution that integrates baseband and radio for intensive BLE and Zigbee applications. Figure 1 and Figure 2 show the layout of the Evaluation Kit (EVK) daughterboards. The layout guide is separated into three functions:

- 1) PCB specifications
- 2) RF layout guidelines
- 3) Power / GND layout guidelines

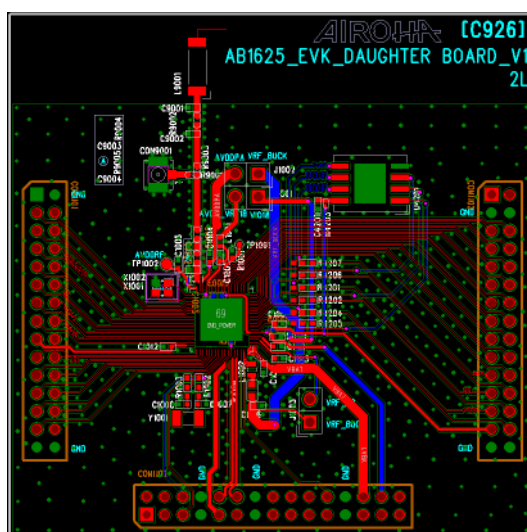


Figure 1. Top View of the AB1625 EVK Daughterboard

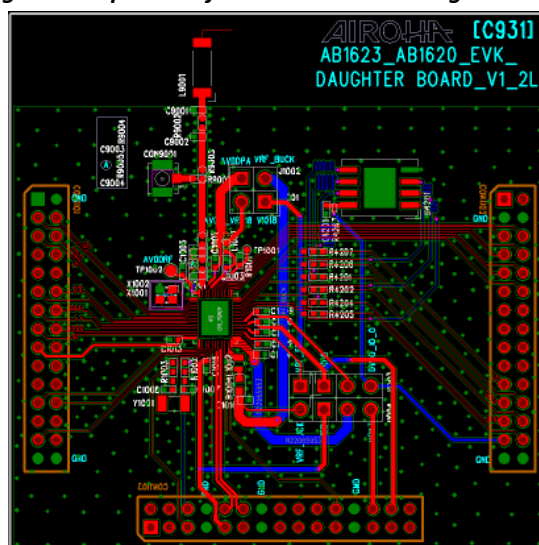


Figure 2. Top View of the AB1623 / AB1620 EVK Daughterboard

2 PCB Specifications

The AB1625 / AB1623 / AB1620 EVK daughterboard has two different layers, as shown in the stack-up layout in Table 1. Changes to the stack-up layers can be made based on the requirements. Impedance must be calculated wire-wide and with a separation. Refer to Table 2 for more details.

Table 1. Stack-up Table of the AB1625 / AB1623 / AB1620 EVK Daughterboard

Top-Side Solder Mask				0.8	mils
L1	TOP	Differential and Signal	copper and plating	1.4	mils
			core	57.75	mils
L2	Bottom	Differential and Signal	copper and plating	1.4	mils
Bottom-side solder mask				0.8	mils
TOTAL				62.15	mils
				1.58	mm

Total thickness: 1.58mm

Table 2. PCB Design Rules of the AB1625 / AB1623 / AB1620 EVK Daughterboard

Parameter	Value	Comments
Number of layers	2	THV
Thickness	1.58 mm	N/A
Dielectric	Er=4.4	Er@1 GHz
Minimum trace width	4.0 mils	N/A
Minimum spacing	4.0 mils	N/A
Laser drill diameter	N/A	N/A
Middle drill diameter	9.84/17.72 mils	N/A
Copper thickness	1 oz	N/A
RF Impedance	Trace width: 28 mil Gnd gap: 6 mil	50 ohm impedance Gnd reference L2
USB Impedance	Trace width: 12.5 mil Trace gap: 6 mil Gnd gap: 6 mil	90 ohm impedance Gnd reference L2

3 RF Layout Guidelines

3.1 RF Trace

A correct layout for the RF section, as shown in Figure 3, is essential for achieving optimum performance of the wireless device. Poor layouts can lead to degradation in output power, harmonic emission, sensitivity, and spectral mask. The PI network for antenna matching should be placed close to the antenna.

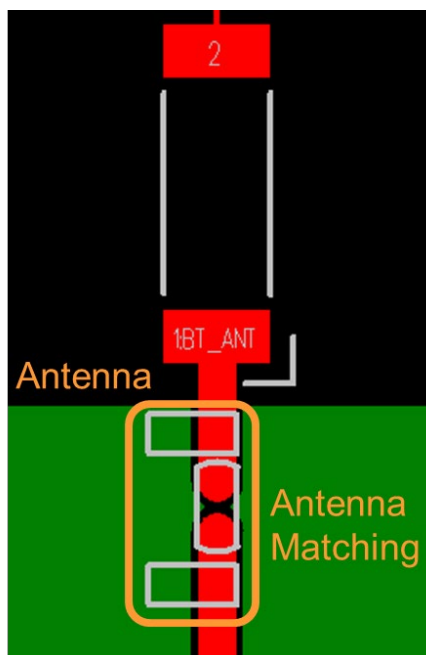


Figure 3. RF Trace of the AB1625 / AB1623 / AB1620 EVK Daughterboard

- Ensure that the RF path impedance is 50 ohm.
- The RF trace to the antenna should be placed on the top layer. Avoid routing it through vias.
- Ensure the ground plane under the RF trace is solid. Increase the number of vias around the RF trace.
- Ensure that the return path is solid and reliable. Refer to Figure 4 for additional information.

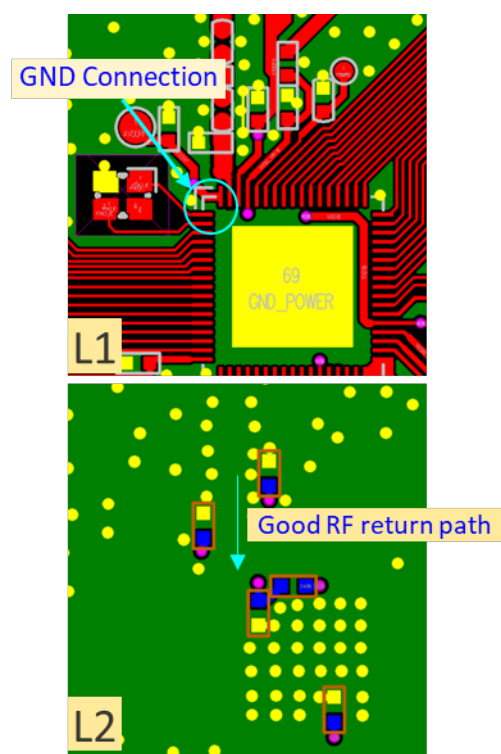


Figure 4. GND Via around RF Trace and Return RF Trace of the AB1625 / AB1623 / AB1620 EVK Daughterboard

3.2 RF Circuit Decoupling Capacitor

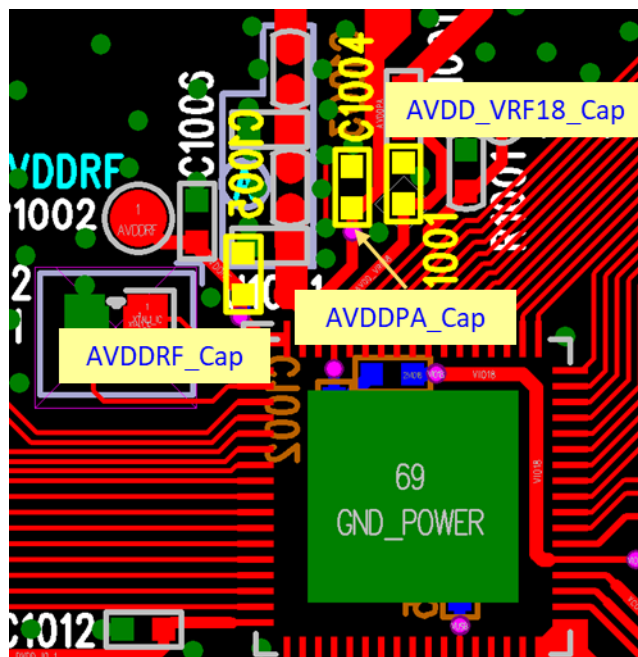


Figure 5. RF Trace of the AB1625 / AB1623 / AB1620 EVK Daughterboard

All decoupling capacitors for AVDD_VRF18 (1uF), AVDDPA (1uF), and AVDDRF (0.47uF) should be placed as close as possible to the pins and with enough GND via for a better return path.

3.3 Crystal 32M

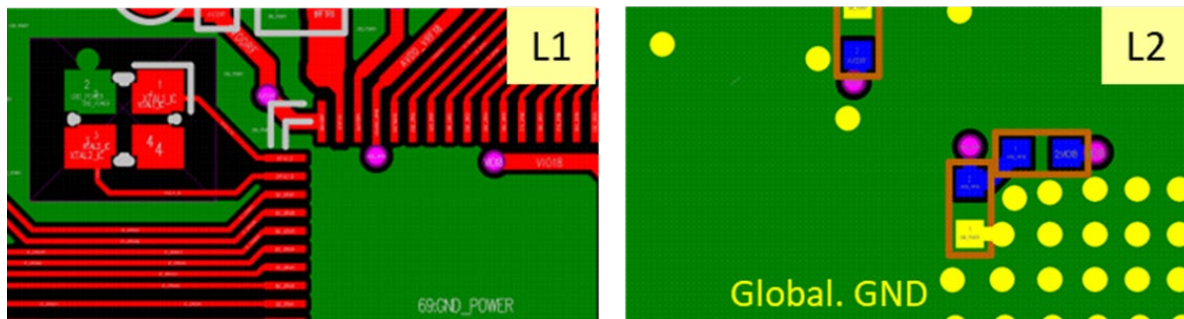


Figure 6. The 32 MHz Crystal on the AB1625 / AB1623 / AB1620 EVK Daughterboard

- The crystal and its trace directly refer to the global ground. Refer to Figure 6 and keep out L1 ground.
- XTAL pins must be put at the top layer if possible. If it is not possible, put them at the bottom layer with the shortest trace to crystal.
- XTAL ground pad directly connects to the global ground. Do not connect to GND on a different layer.
- Increase the number of GND vias surrounding the crystal to enhance isolation.

4 Power Layout Guidelines

4.1 Power Trace

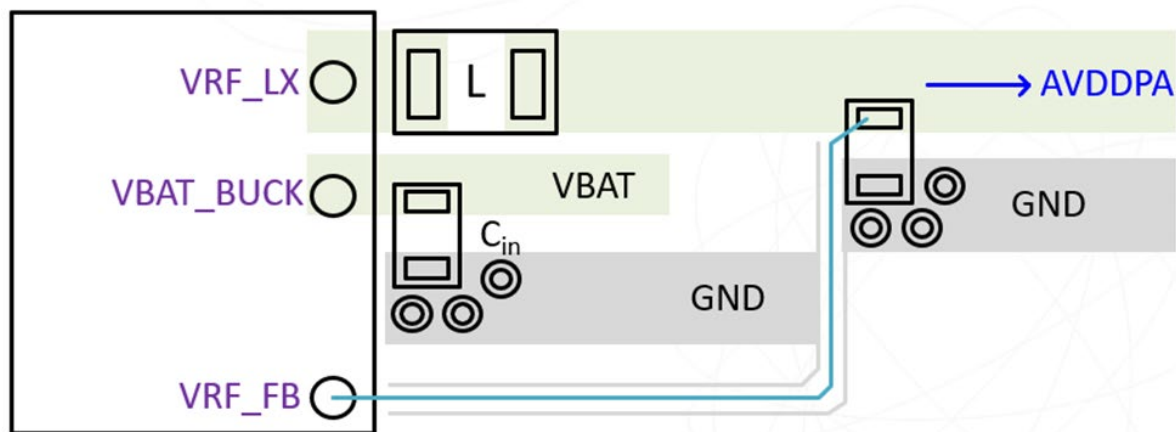


Figure 7. Buck Layout Guide

- The BUCK input capacitor must be placed as close as possible to the VBAT_BUCK pin.
- The BUCK inductor must be placed as close as possible to the VRF_LX pin.
- BUCK input cap GND connects with main ground directly as shown in Figure 8.
- BUCK input cap must be the nearest component, followed by the BUCK inductor.
- The sensing point of VRF_FB path must be located at the maximum output capacitor.
- VRF_FB must be shielded with GND and have no noise above or below it.

4.2 Buck Trace

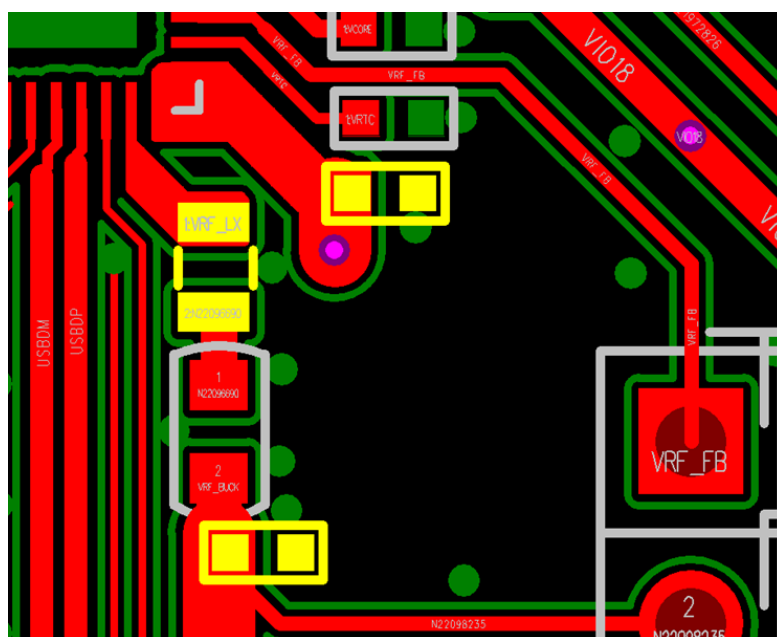


Figure 8. Placement and the LX Trace on the AB162x EVK Daughterboard

The VBAT, VBAT_BUCK, and VRF must be kept as wide as possible. Refer to Table 3 for additional information.

Table 3. Minimum Requirements for Power Trace Width on AB162x

Parameter	Trace Width (mil)	GND Shielding
VBAT	12	
VBAT_BUCK	12	
VRF_FB	6	YES
VRF_LX	10	
VCORE	6	
VRTC	4	
VIO18	6	

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